

folgt aus reeller Tal

$$a: \mathbb{N} \rightarrow \mathbb{R}$$

$$\{a\}_n = a_1, a_2, a_3$$

V: skiver $a(n) = a_n$

$$(a_n)_n$$

$$X_n = X_n^2 + i$$

$$m \leq i \leq n$$

Rahner

$$\sum_{i=1}^n a_i$$

geometrisch folge

$$c=1 \quad r=\frac{1}{2}$$

$$f_1 = 1 \quad f_2 = 1 \cdot \frac{1}{2}$$

$$f_{n+1} = f_n \cdot r$$

Geometrisch verfahren

$$\sum_{i=0}^n \left(\frac{1}{2}\right)^i = 2 - \left(\frac{1}{2}\right)^n$$

$$\sum_{i=0}^{\infty} r^i = \begin{cases} \frac{r^{n+1}-1}{r-1} & r \in \mathbb{R} \setminus \{1\} \\ n+1 & r = 1 \end{cases}$$

$$S = \sum_{i=0}^n r_i$$

$$r \cdot S = \sum_{i=0}^{i=n} r^{i+1} = \sum_{i=1}^{n+1} r^i = S + r^{n+1} - 1$$

$$r \cdot S - S = r^{n+1} - 1$$

$$S = \frac{r^{n+1} - 1}{r - 1}$$

Aritmetisch Folge

$$b=1 \quad d=2$$

$$f_1 = 1 \quad f_2 = 3$$

$$f_{n+1} = f_n + d$$

$$\sum_{i=0}^n i = \frac{n(n+1)}{2}$$

$$\text{Wie } \frac{(n+1)q}{2} \quad \text{also } \frac{(n-1)}{2} \cdot (n+1) + \frac{(n+1)}{2} = \frac{n(n+1)}{2}$$

$$\sum_{i=0}^n (b + id) = \sum_{i=0}^n b + d \cdot \sum_{i=0}^n i$$

$$\sum_{i=0}^n b = b \sum_{i=0}^n 1 = b(n+1) + d.$$

$$\sum_{i=0}^n (b+id) = b(n+1) + d \cdot \frac{n(n+1)}{2}$$